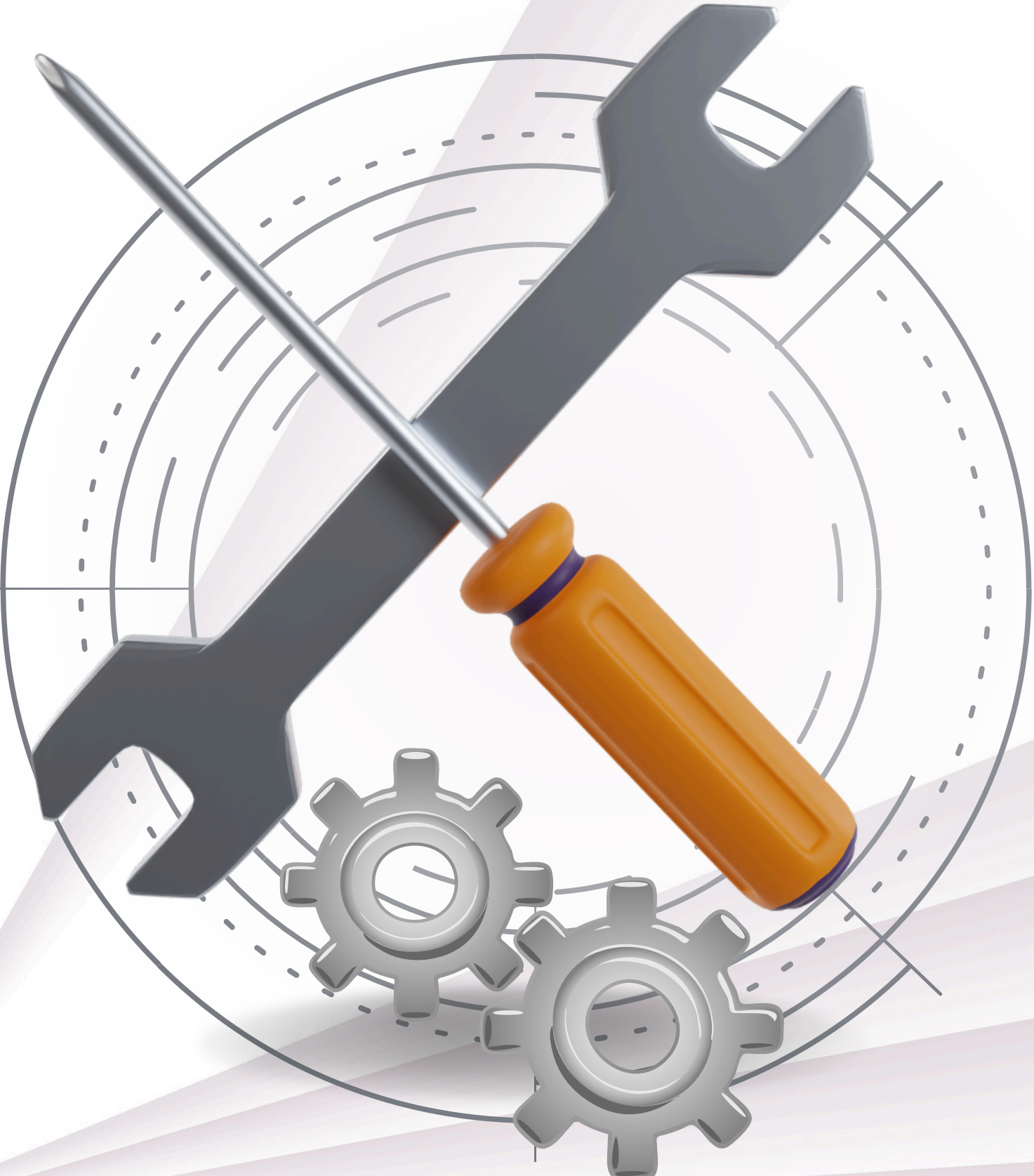
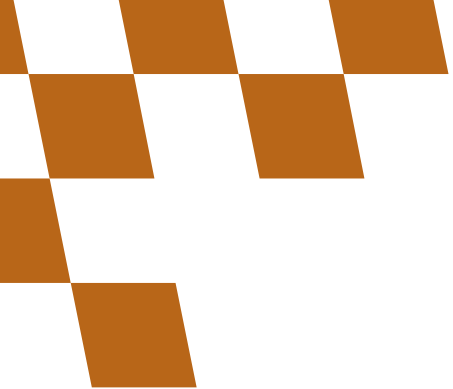
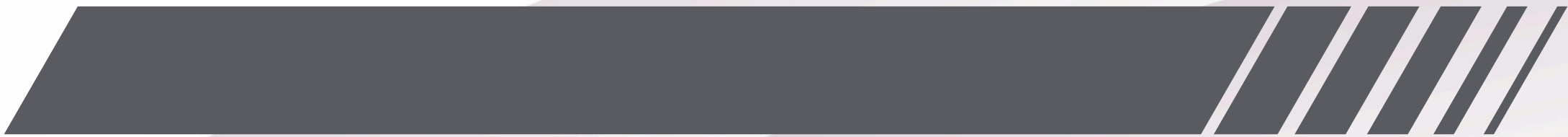




Predictive Maintenance for Industrial Machinery



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 - Maisoun Ebrahim Ahmed
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 - Fatma Saber Elgendy
- 

Problem & Objectives

The Problem:

Unexpected machine failures lead to production downtime and high maintenance costs.

Objectives:

- Predict failures in advance using sensor data (vibration, temperature, pressure).
- Reduce maintenance costs and extend equipment lifespan.



Why is it important problem ?

- Stops machines from breaking down suddenly.
- Keeps production running without stopping.
- Saves continuous repair costs.

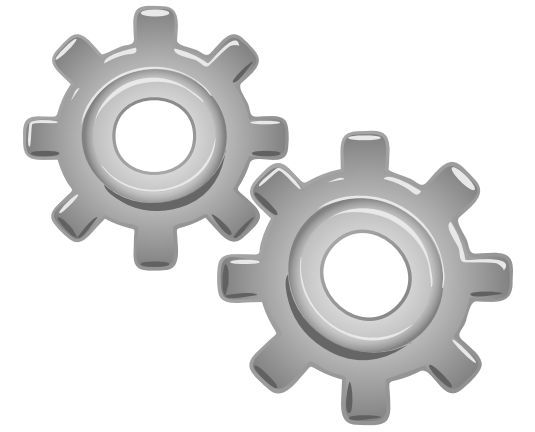


What is your proposed ML solution?

- Analyze machine operating data (Temperature, Rotational Speed, Torque, and Tool Wear) to detect potential failures early.
- Build a Predictive Maintenance model using Machine Learning.
- Keeps production running without stopping.
- Saves continuous repair costs.

BUSINESS

MODEL CANVAS



Designed for : Smart Factories & Industrial Enterprises

KEY PARTNERS

IoT Hardware Vendors

Cloud Infrastructure Providers

Industrial Integration & Consulting Firms

KEY ACTIVITIES

Data Preprocessing & Feature Engineering

Model Development & Tuning

API Development

VALUE PROPOSITION

Predictive Failure Detection

Downtime & Cost Reduction

CUSTOMER RELATIONSHIPS

Dedicated Technical Support

Onboarding & Training

Service Level Agreements (SLA)

CUSTOMER SEGMENTS

Manufacturing & Heavy Industrial Plants

Operations & Maintenance Departments

KEY RESOURCES

AI Models & Software

Expert Human Capital

Computing Infrastructure

Actionable Operational Insights

CHANNELS

Direct B2B Enterprise Sales

Industrial & AI Trad Shows

System Integrator Networks

COST STRUCTURE

Cloud & Server Infrastructure Costs

Marketing & Customer Acquisition Costs

R&D & Engineering Salaries

REVENUE STREAMS

Software as a Service (SaaS) Subscriptions

Maintenance & Model Re-training Contracts

Setup & Integration Fees

ML Task & Model Prediction

- Type: Supervised Learning.
- Classification: Predict Machine Health Status (0 = Normal, 1 = Faulty).



Original Dataset

	UDI	Product ID	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Target
0	1	M14860	M	298.1	308.6	1551	42.8	0	0
1	2	L47181	L	298.2	308.7	1408	46.3	3	0
2	3	L47182	L	298.1	308.5	1498	49.4	5	0
3	4	L47183	L	298.2	308.6	1433	39.5	7	0
4	5	L47184	L	298.2	308.7	1408	40.0	9	0

Before Feature Selection & Engineering

Final Feature Dataset

	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Target	Temp_Diff	Power_W	Overstrain	Torque_Speed_Ratio
0	1.0	298.1	308.6	1551	42.8	0	0	10.5	6951.590560	0.0	0.027595
1	0.0	298.2	308.7	1408	46.3	3	0	10.5	6826.722724	138.9	0.032884
2	0.0	298.1	308.5	1498	49.4	5	0	10.4	7749.387543	247.0	0.032977
3	0.0	298.2	308.6	1433	39.5	7	0	10.4	5927.504659	276.5	0.027565
4	0.0	298.2	308.7	1408	40.0	9	0	10.5	5897.816608	360.0	0.028409

After Removing Non-Predictive Columns & Adding Engineered Features

Visualization





Welcome to Predictive Maintenance Dashboard

Explore machine performance, failure patterns, and maintenance insights through interactive analysis.

Visualization

Executive Overview



Failure Analysis





Executive Overview

Total Records
10K



Total Failures
339



Failure Rate
3.39%



Failure Free Rate
96.61%



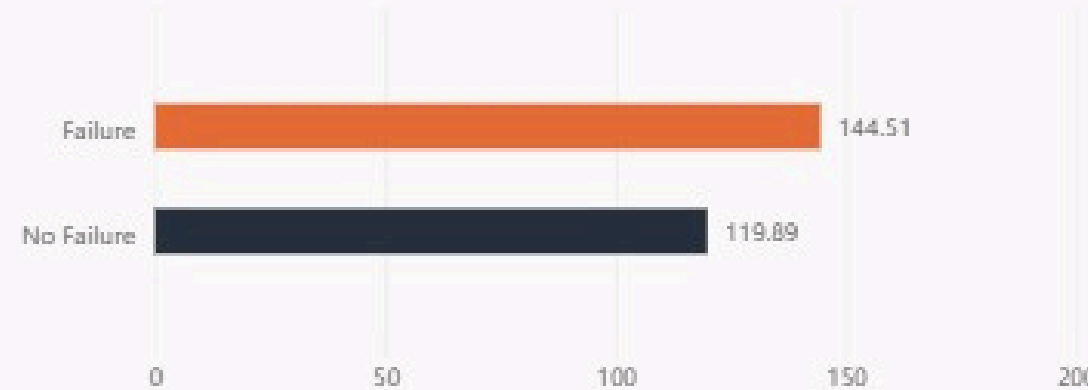
Average Tool wear
107.95



Failure vs No Failure



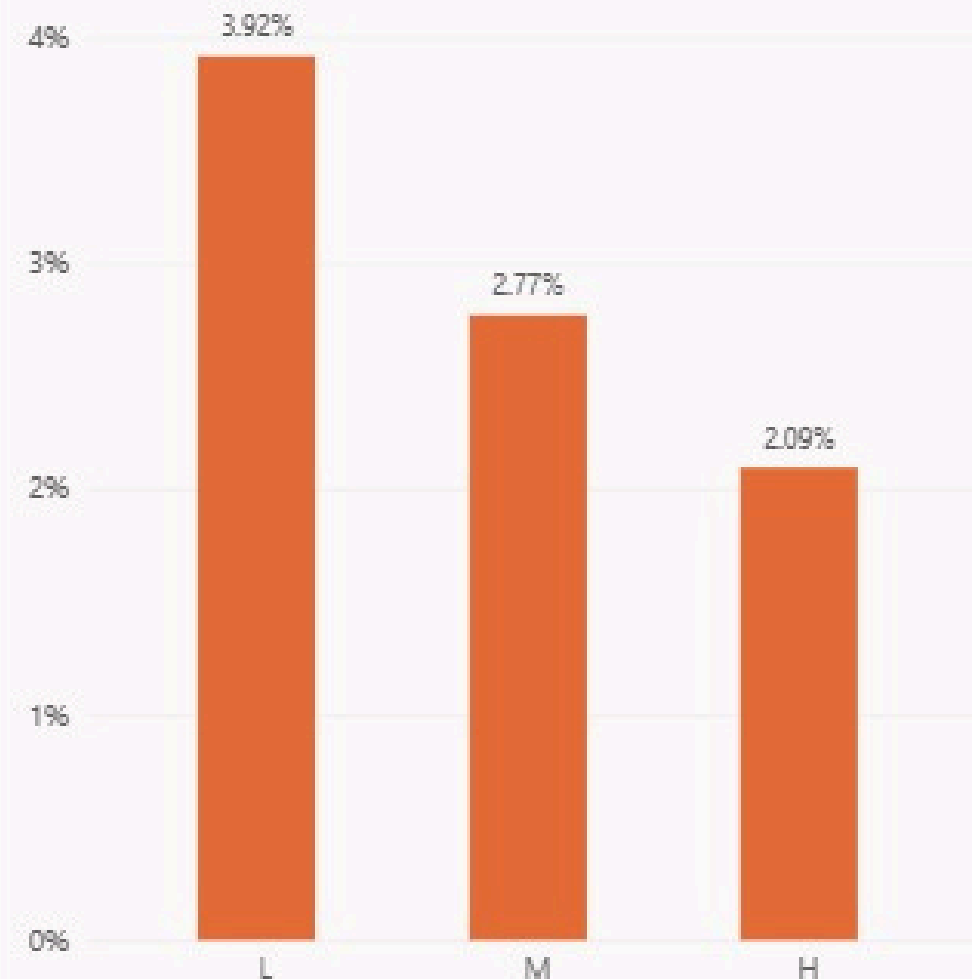
Average Tool Wear



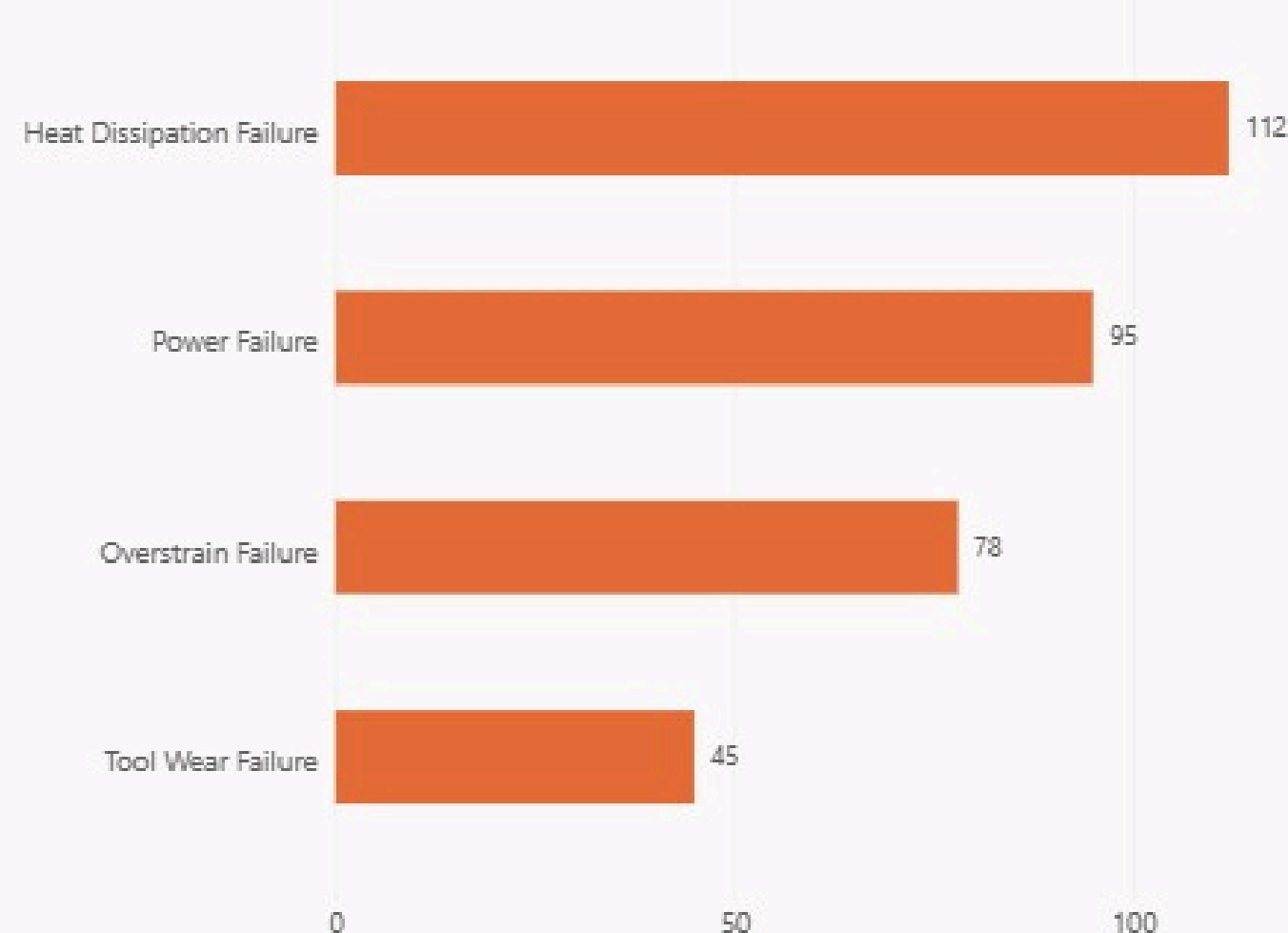
Average Torque



Failure Rate by Machine Type



Failure Type Distribution



Operational Profile

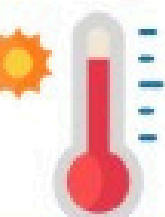
Average Torque
39.99



Average RPM
15M



Average Process temperature
310.01



Machine Type

All

Failure Type

All



Failure Analysis

Total Failures

339



Failure Rate

3.39%



Failure Free Rate

96.61%



Top Failure Type

Heat Dissipation Failure



Failure Types Count

5

Machine Type

All

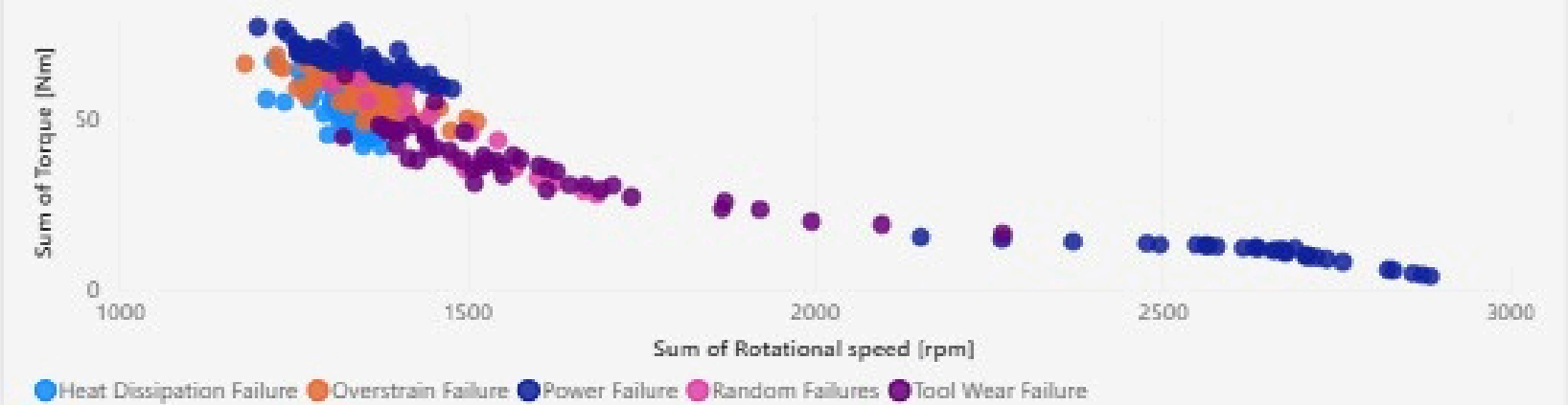
Failure Type

All

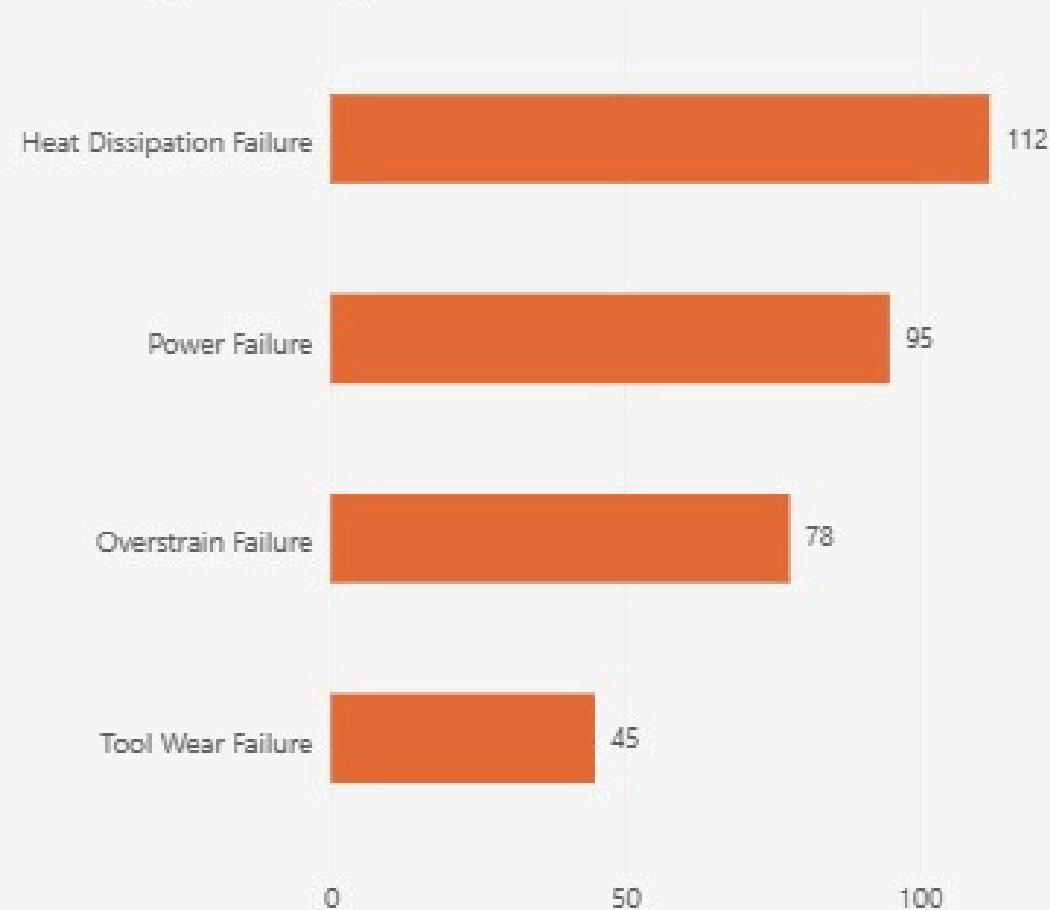
Failure Type × Machine Type

Type	Heat Dissipation Failure	Overstrain Failure	Power Failure	Tool Wear Failure	Total
H	8	1	5	6	20
L	74	73	59	25	231
M	30	4	31	14	79
Total	112	78	95	45	330

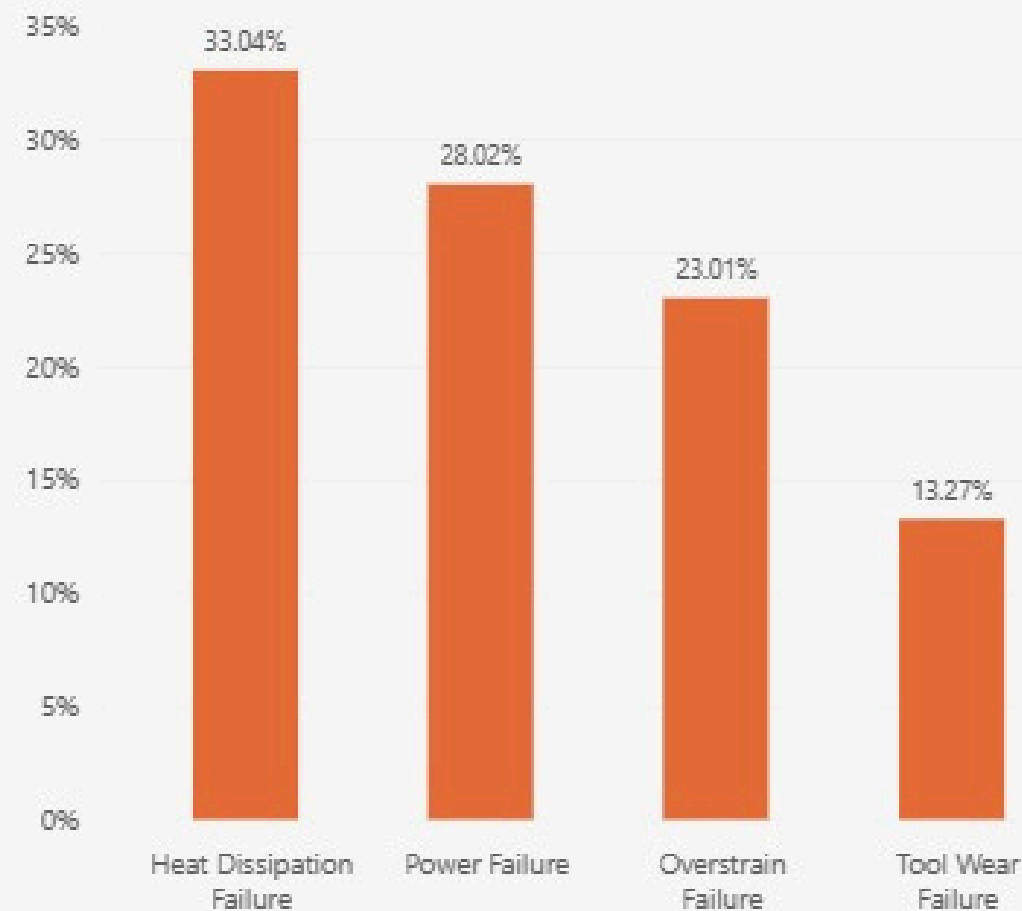
Torque vs RPM Failure Pattern



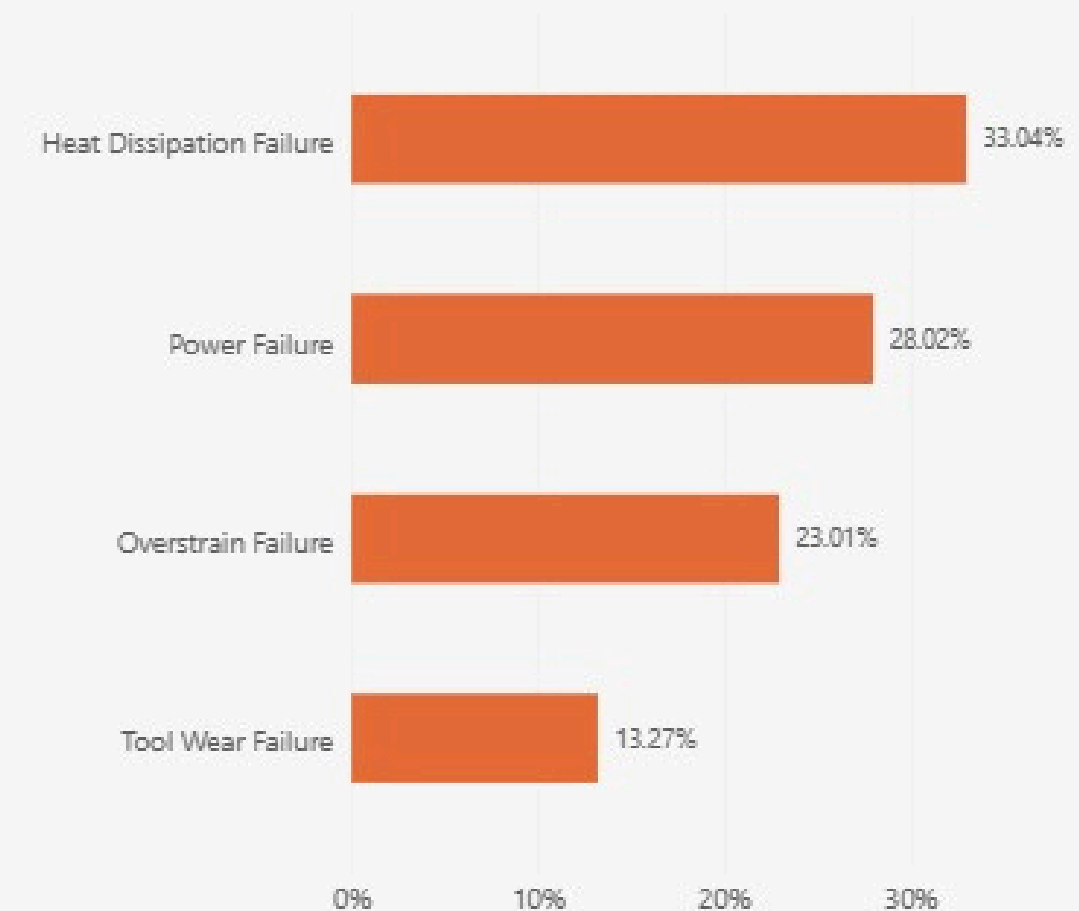
Failure Type Ranking



Failure Type Distribution



Failure Contribution



Exploratory Data Analysis (EDA)

- **Summary Statistics:**

Check dataset shape (10,000Rows &10 Columns), data types, and basic statistics (Mean, Std, Min, Max).

- **Target Class Distribution (Class Imbalance):**

Check the proportion of Normal vs. Failure states (Detect highly imbalanced failure events).

Target	
0	9661
1	339

- **Correlation Analysis:**

Heatmap to understand relationship between sensor readings (e.g., Torque vs. Rotational Speed).

- **Outlier Detection & Distributions:**

Boxplots and Histograms to inspect feature distributions and abnormal sensor spikes.

Data Cleaning

- **Data Verification:**
Checked dataset completeness (0 Missing Values, 0 Duplicate Records).
- **Feature Dropping:**
Removed non-predictive identifier columns (UDI and Product ID).
- **Noise Inspection:**
Verified sensor reading ranges to ensure no unrealistic measurement errors exist.

Preprocessing

- **Encoding:**

```
en = OrdinalEncoder(categories=[['L', 'M', 'H']])  
df[['Type']] = en.fit_transform(df[['Type']])
```

- **Scalling:**

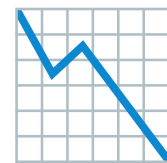
```
scaler = RobustScaler()  
X_train = scaler.fit_transform(X_train)  
X_test = scaler.transform(X_test)
```

- **Balanceing:**

```
smote = SMOTE(sampling_strategy=0.3 , random_state=42)  
X_train, y_train = smote.fit_resample(X_train, y_train)
```

Baseline Model

Logistic Regression



- Purpose:
Build a simple model to establish a minimum performance benchmark.

```
===== Test =====
accuracy_score : 0.942

Confusion matrix :
[[1834   98]
 [  18   50]]

Classification Report :
              precision    recall  f1-score   support

      0               0.99      0.95      0.97       1932
      1               0.34      0.74      0.46         68

   accuracy               0.94       2000
  macro avg               0.66      0.84      0.72       2000
 weighted avg             0.97      0.94      0.95       2000
```

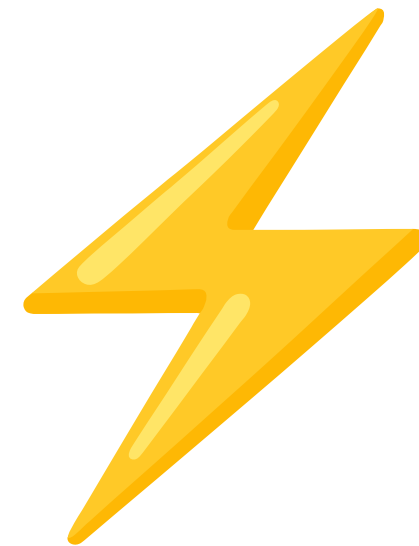
Try multiple model



Decision Tree



Random Forest



XGBoost





Decision Tree

Recall is Key:

Focused on maximizing Recall for Failure Class (1) to catch all potential breakdown events.

```
===== Test =====  
accuracy_score : 0.977
```

```
Confusion matrix :
```

```
[[1901  31]  
 [ 15  53]]
```

```
Classification Report :  
precision
```

	precision	recall	f1-score	support
0	0.99	0.98	0.99	1932
1	0.63	0.78	0.70	68
accuracy			0.98	2000
macro avg	0.81	0.88	0.84	2000
weighted avg	0.98	0.98	0.98	2000



Random Forest



IS

Higher Accuracy & Reliability

===== Test =====
accuracy_score : 0.9885

Confusion matrix :
[[1922 10]
[13 55]]

Classification Report :

	precision	recall	f1-score	support
0	0.99	0.99	0.99	1932
1	0.85	0.81	0.83	68
accuracy			0.99	2000
macro avg	0.92	0.90	0.91	2000
weighted avg	0.99	0.99	0.99	2000

XGBoost

- **Equal High Recall:** Both models achieved the same maximum Recall, capturing all potential machine failures.
- **Superior Precision:** XGBoost significantly improved Precision, drastically reducing False Alarms.
- **Optimal Balance:** Achieved a higher overall F1-Score, making it the most cost-effective solution for deployment.

	Model A	Model B
Recall	1.00	1.00
Precision	0.85	0.95
F1-Score	0.92	0.98

Model Comparison

Model	Threshold	Test Precision	Test Recall	Test F1-Score	Test Accuracy
Logistic Regression	0.5	33.80%	73.50%	46.30%	94.20%
Decision Tree	0.7	63.10%	77.90%	69.70%	97.70%
Random Forest	0.7	84.60%	80.90%	82.70%	98.80%
XGBoost (Winner)	0.7	87.30%	80.90%	84.00%	99.00%

Detailed Model Comparison

	Best Threshold	Train Accuracy	Test Accuracy	Train Precision	Test Precision	Train Recall	Test Recall	Train F1 Score	Test F1 Score	Train Balanced Accuracy	Test Balanced Accuracy
Model											
Logistic Regression	0.5	0.879168	0.9420	0.786010	0.337838	0.654443	0.735294	0.714218	0.462963	0.800504	0.842285
Decision Tree	0.7	0.970041	0.9770	0.972808	0.630952	0.895168	0.779412	0.932375	0.697368	0.943832	0.881683
Random Forest	0.7	0.983677	0.9885	0.986450	0.846154	0.942192	0.808824	0.963813	0.827068	0.969155	0.901824
XGBoost	0.7	0.999403	0.9895	0.998706	0.873016	0.998706	0.808824	0.998706	0.839695	0.999159	0.902341

Real-World Interpretation

- **Deployment Choice:** XGBoost selected as the optimal production model.
- **Proactive Prevention:** Detects 80.9% of failure events early.
- **Operational Efficiency:** High 87.3% Precision minimizes false alarms.
- **Business Impact:** Cuts downtime and reduces unnecessary repair costs.



***Any
Question?***